

Sudden Death Due to Cerebral Leukemic Hemorrhage Occurring After Acupuncture Treatment for Gingival Bleeding

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Abstract: A 45-year-old woman who experienced stomalgia and gingival bleeding for several days died unexpectedly after acupuncture treatment. At autopsy, trivial injuries on the liver and the stomach and mild hemoperitoneum due to improper acupuncture were found. Also, acute lymphoblastic leukemia and hyperleukocytosis were diagnosed by postmortem examinations. Intracranial hemorrhage due to undiagnosed acute lymphoblastic leukemia was identified as the cause of death. Moreover, the relationship between therapeutic misadventure and death was also determined. We suggest that undiagnosed leukemia should be considered as a differential diagnosis when sudden death occurs owing to intracranial hemorrhage. If therapeutic misadventure was involved, it is also of great importance to assess the relationship between that and death in forensic expertise.

Key Words: leukemia, acupuncture, intracranial hemorrhage, therapeutic misadventure

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Leukemia is characterized by potential hemorrhagic complications, including purpura, gingival bleeding, and intracranial hemorrhage among others. Fatal intracranial hemorrhage occurs suddenly and is often the cause of unexpected death in leukemic patients.^{1–3} Hyperleukocytosis caused by acute lymphoblastic leukemia (ALL) is associated with early mortality owing to the hyperviscosity created by the high leukocyte count. Patients with hyperleukocytosis are at risk for unexpected death due to intracranial hemorrhage^{4–6}; however, it seems to occur much less than before.^{5,7}

Here, we reported the first autopsy case of a woman who died of intracranial hemorrhage due to undiagnosed ALL in which medical malpractice attributed to improper acupuncture was also involved.

CASE REPORT

A 45-year-old woman was admitted into a dental clinic at 8 PM for stomalgia and gingival bleeding. Except for the intra-

venous infusion of glucose-saline, Arilin, and albomycin, the dentist performed acupuncture on several acupoints for half an hour, including Guanyuan (CV4), Shuifen (CV9), Zhongwan (CV12), and Xiawan (CV10). The 4 acupoints are located on the anterior median line of the abdomen. Zhongwan (CV12), Xiawan (CV10), and Shuifen (CV9) are located approximately 13.3, 6.66, and 3.33 cm above the navel respectively, whereas Guanyuan (CV4) is located 9.99 cm below the navel. All should be acupunctured vertically by needles to a depth⁸ of approximately 1.66 to 3.33 cm. The woman vomited twice during acupuncture, and subcutaneous hemorrhage was noticed at the acupoints. She was sent back home at 11 PM and found dead at 10 AM the next morning. No hematologic profile was available.

Autopsy Findings and Histopathologic Examination

Forensic autopsy was performed 4 days after death. Subcutaneous hemorrhages were found on the back and the extremities. Also, patchy subcutaneous hemorrhagic sites were present on the abdomen, located approximately 3.0 cm below the xiphoid process (location of Zhongwan [CV12] and Xiawan [CV10]), 1.7 cm above the navel (location of Shuifen [CV9]), and 7.0 cm above the pubic symphysis (location of Guanyuan [CV4]).

A rupture was found on the right temporal lobe. Examination of sections showed intracerebral hemorrhage in the right temporal lobe and the occipital lobe (Fig. 1), which had reached into the right lateral ventricle and the right middle cranial fossa via the rupture on the right temporal lobe, resulting in a subdural hematoma (weighting approximately 21.5 g) composed of numerous leukemia cells and few erythrocytes. Multiple leukemic nodules composed of leukemia cells and erythrocytes were found in the right temporal lobe and the occipital lobe (Fig. 2). Approximately 500 mL of blood was found in the abdominal cavity. Interestingly, a blood clot was discovered on the left lobe of the liver. After removing the clot, a fine rupture with numerous superficial and parallel scratches was found (Fig. 3). The clot was a mixed thrombus composed of many leukemia cells and erythrocytes and a few strands of fibrin and platelets. Absence of the liver capsule was also observed at the rupture sites and scratches, accompanied by the exudation of leukemia cells and fibrins. The stomach was also involved, presenting as hemorrhage and infiltration of leukemia cells in all layers. Moreover, leukemia cells aggregated in vessels or exudated and infiltrated nonhemopoietic tissues, including the brain, the lungs, the spleen, and other organs.

Cytological Examination

Bone marrow smears showed severe autolysis. Postmortem examination of a hemogram showed that the white blood cell (WBC) and lymphocyte counts were $320 \times 10^9/L$ and $294.4 \times 10^9/L$, respectively. Because of hemolysis, the red blood cell and platelet counts remained unknown. Numerous leukemia cells and very few red blood cells were found on the blood smear.

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FIGURE 1. Intracerebral hemorrhage in the right temporal lobe and occipital lobe, reaching into the right lateral ventricle and the rupture on the right temporal lobe.

Wright staining of blood smear showed that leukemia cells and nuclei were circlelike, with an increase in nuclear-cytoplasmic ratio. Also, 2 or 3 nucleoli were noticed (Fig. 4). Because of severe autolysis, cytochemical and immunohistochemical stainings for classification of leukemia had negative results; nevertheless, according to the features of leukemia cells, ALL was diagnosed.

Cause of Death

The cause of death was intracranial hemorrhage due to undiagnosed ALL.

DISCUSSION

Since 1965, a number of serious or even life-threatening complications of acupuncture have appeared. The most frequently reported complications were pneumothorax,^{9,10} lesions of the spinal cord,^{11,12} and infection.^{13,14} The injuries described in this article could have been prevented if the practitioners had proper anatomical knowledge and were more skilled. The woman was acupunctured for half an hour, aiming at hemostasis and pain relief. Interestingly, trivial stabs on the liver and the stomach, hemoperitoneum, and subcutaneous hemorrhages at the sites of the acupoints were found at autopsy. It was confirmed that the needles had perforated the abdominal wall when the acupoints Zhongwan (CV12) and Xiawan (CV10) were stimulated, and the liver and the stomach were stabbed by the needles,

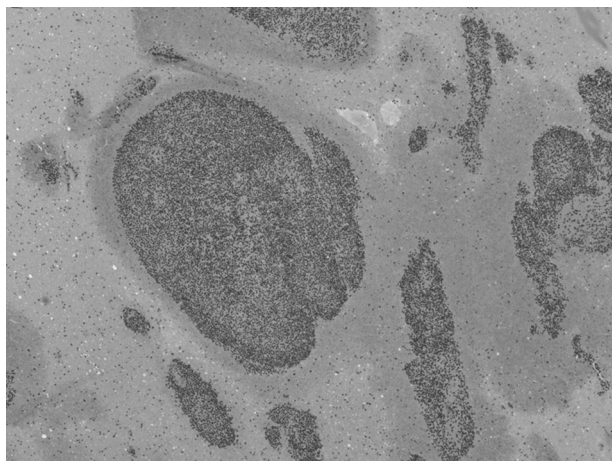


FIGURE 2. Leukemic nodules in the right temporal lobe (hematoxylin and eosin; original magnification $\times 100$).

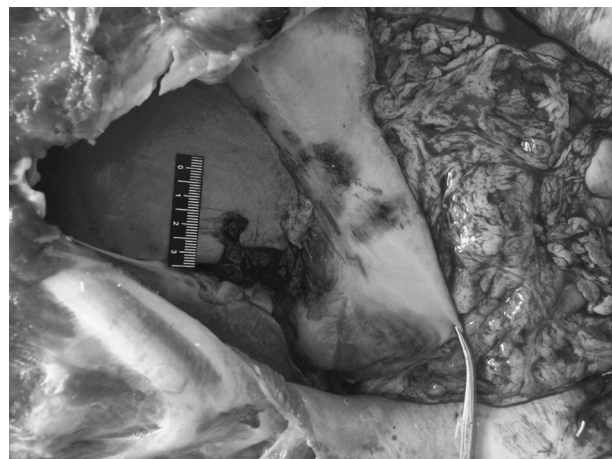


FIGURE 3. A mixed thrombus adhered to the fine rupture on the left lobe of the liver, accompanied by many superficial and parallel scratches; patch hemorrhage on the stomach; hemoperitoneum.

resulting in hemorrhage and formation of thrombus on the left lobe of the liver, hemorrhage in the stomach, and hemoperitoneum. Also, the tips of the needles brought approximately much abrasions around the stabbed areas on the liver during the movement of the diaphragm and the liver while breathing. However, although the acupuncture procedure left some confounding findings, such as local hemorrhage, it was not thought to be the cause of death and could not directly be linked to the cerebral hemorrhage.

Hyperleukocytosis was diagnosed by postmortem examinations, showing the significant increase of leukemia cells in vessels and extensive infiltration around the vessels in the brain, which could explain the decedent's bleeding tendency.^{15,16} Nowacki et al¹⁷ pointed out that a WBC count higher than $100 \times 10^9/L$ was the critical point of spontaneous intracranial hemorrhage. Leukostasis caused by hyperleukocytosis is the interference of blood flow by WBCs within the microcirculation.⁵ It primarily involves cerebral microvasculature and is characterized by the stasis of leukemia cells within vessels, resulting in the compression, invasion, and infiltration of vascular walls, which

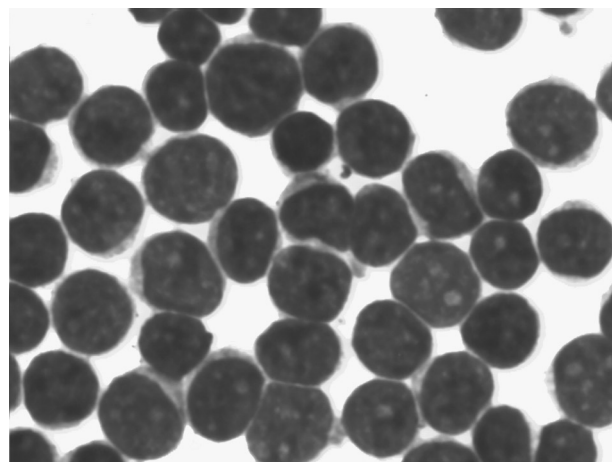


FIGURE 4. Immature lymphocytes with a higher nuclear-cytoplasmic ratio and multiple nucleoli (Wright staining; original magnification $\times 1000$).

give rise to microcirculation disturbance and hypoxic injury of endothelial cells and angiorrhexis.¹⁸ Hence, the following neuropathologic changes were present in this case: cellular exudation into the Virchow-Robin space, appearance of leukemic nodules (admixture of leukemia cells and erythrocytes), and mechanical compression of the capillaries and the venules by the enlarging leukemic nodules.

Undiagnosed leukemia should be considered as a differential diagnosis when sudden death occurs owing to intracranial hemorrhage. If therapeutic misadventures are suspected during forensic expertise, it is very important to determine its relationship with the cause of death.

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